Thermal Burn

TB-09 | Trauma & Burns

Quick flow

01 Stop Burning [EMT / AEMT / Paramedic]

Remove the heat source and constricting items; protect responders. For a small thermal burn (<10% TBSA), use cool running water for approximately 20 minutes when feasible without hypothermia or transport delay. Monitor temperature and keep uninjured areas warm; no ice.

02 Airway / Trauma Threat? [EMT / AEMT / Paramedic]

Assess inhalation injury, progressive respiratory compromise, CO/cyanide risk, and multisystem trauma; plan early airway care when clinically indicated.

03 Classify + Estimate TBSA [EMT / AEMT / Paramedic]

Count only partial- and full-thickness burns. For scattered burns, the patient's palm plus fingers (whole palmar hand surface) is approximately 1% TBSA; use an age-adjusted validated chart for larger pediatric burns.

04 Burn-Center Consultation? [EMT / AEMT / Paramedic]

Full thickness • partial thickness ≥10% TBSA • potentially deep burn of any size • critical-area burn • inhalation injury • associated trauma • uncontrolled pain • complicating comorbidity • pediatric burn.

05 Dry Cover + Resuscitate [AEMT / Paramedic]

Prevent hypothermia • IV/IO when indicated • controlled warmed LR infusion using the burn-fluid table below (NS if LR unavailable) • pain control • frequent reassessment. Do not give an automatic 1 L bolus solely for burn size; shock requires assessment for hemorrhage or another cause.

06 Appropriate Destination [EMT / AEMT / Paramedic]

Consult a burn center for indicated injuries, including potentially deep burns of any size and complicating comorbidity. Consultation does not automatically require bypassing needed stabilization; follow the regional destination plan. Use a closer stabilization hospital for an immediate critical need that cannot safely be managed during longer transport; notify early, document the reason, and coordinate transfer.

Overview

- Stop the burning process, identify airway or associated trauma, estimate partial- and full-thickness TBSA, and prevent hypothermia.
- Full-thickness burns; partial-thickness burns of 10% TBSA or greater; deep burns involving critical areas; inhalation injury; significant associated trauma; uncontrolled pain; and pediatric burns require burn-center consultation or transfer consideration.

Provider-Level Actions

Perform all immediately indicated care within scope without delaying rapid trauma transport.

EMT

- Stop the burning process and remove jewelry/constricting items. For a small thermal burn (<10% TBSA), use cool running water for approximately 20 minutes when feasible without hypothermia or transport delay; monitor temperature and keep uninjured areas warm. Do not cool when hypothermia risk is significant. Cover with clean dry dressings, prevent hypothermia, and estimate TBSA.
- Support airway and ventilation, screen for carbon monoxide/cyanide exposure, and treat associated trauma.

AEMT

- Perform all EMT care plus IV/IO access and authorized non-IM analgesia within scope. When burn-fluid resuscitation is indicated, use a controlled warmed lactated Ringer's infusion (normal saline if LR is unavailable) at the starting rate in the burn-fluid table; reassess perfusion, respiratory status, and response frequently.
- Expanded AEMT analgesia is not enabled by default while individual authorization is unverified. The specified medication, route, and patient group require current Tennessee authority, documented Tennessee-approved expanded analgesia training, agency credentialing, and medical-director standing orders; training alone does not grant permission.
- Consider two IV sites for larger burns without delaying transport.
- For shock, search for associated hemorrhage or another cause and follow the appropriate shock-resuscitation pathway; the isolated-burn starting rate does not replace shock treatment.

Paramedic

- Perform all prior care plus advanced airway management when clinically required, protocol-directed analgesia, and burn resuscitation.
- Coordinate burn-center consultation or direct transport when appropriate. Consultation does not automatically require bypassing needed stabilization; apply the regional destination plan, document any immediate critical need for closer stabilization, and arrange transfer.

Indications

- Patient with suspected thermal burn requiring prehospital assessment or treatment.

Assessment | Initial burn-fluid rates when resuscitation is indicated

- Record injury time, weight, and partial-/full-thickness TBSA. Use warmed lactated Ringer's when available; normal saline is an alternative, changing to LR when available.
- TBSA below 20%: 0.25 mL/kg × TBSA percentage per hour. Enter TBSA as a whole percentage, not a decimal fraction; for 70 kg and 15% TBSA, the starting rate is approximately 263 mL/hr.
- TBSA 20% or greater: age ≤5: 125 mL/hr; age 6-14: 250 mL/hr; age ≥15: 500 mL/hr.
- Reassess perfusion, respiratory status, and response frequently. Do not give an automatic 1 L bolus solely because a burn is extensive; do not delay transport.
- For shock, search for hemorrhage or another cause and use the appropriate shock-resuscitation pathway rather than relying on the isolated-burn starting rate.

Assessment | Burn-center consultation or transfer considerations

- Full-thickness burns; potentially deep burns of any size; or comorbidity that complicates care.
- Partial-thickness burns of 10% TBSA or greater.
- Deep partial- or full-thickness burns involving the face, hands, feet, genitalia, perineum, or over major joints.
- Suspected inhalation injury, chemical or electrical injury, significant associated trauma, poorly controlled pain, or pediatric burns requiring specialty support.

Treatment steps

- For a small thermal burn (<10% TBSA), use cool running water for approximately 20 minutes when feasible without hypothermia or transport delay. Monitor temperature and keep uninjured areas warm; do not use ice or cool when hypothermia risk is significant. Then cover with clean dry dressings.
- For an isolated burn requiring fluid resuscitation without shock, begin a controlled warmed LR infusion using the displayed age-/weight-/TBSA-based burn-fluid table; use NS if LR is unavailable. Record injury time, weight, and partial-/full-thickness TBSA, and reassess frequently.

- For shock, search for associated hemorrhage or another cause and use the appropriate shock-resuscitation pathway; do not delay transport. Do not apply medical/septic-shock fluid targets indiscriminately to pediatric traumatic hemorrhage.
- Consult a burn center for indicated injuries, potentially deep burns of any size, or complicating comorbidity. Consultation does not automatically require bypassing needed stabilization; follow the regional destination plan and document/coordinate any closer stabilization and transfer.

Warnings / precautions

- Do not include superficial first-degree burns in TBSA.
- Do not apply ice, wet sheets, creams, or ointments to significant burns; burn patients are highly susceptible to hypothermia.
- Do not use intramuscular analgesics in burn patients.

Related pathways / contacts

AM-05 Hypotension / Shock - Adult shock care when hypotension suggests hemorrhage or another cause beyond an isolated burn.

TB-06 Multiple Trauma - Associated traumatic hemorrhage and trauma destination priorities.

Source and references

Current Tennessee EMS scope of practice, verified individual credentials, and Claiborne County standing orders control provider authorization.

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